Nilesh Pawar

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SUMMARY

Final-year Computer Engineering student focused on AI and Generative AI (GenAI) applications and Python/FastAPI backend development. Hands-on project experience with LLM APIs, RAG, vector search, and tool-calling workflows; knowledge of LangChain and LangGraph.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL
AI & GenAI: LLM API Integration (Gemini, Groq, NVIDIA Nemotron), Prompt Engineering, RAG Fundamentals, Embeddings & Vector Search, LangChain, LangGraph, LangSmith, Ragas, Tool Calling
Backend: FastAPI, Node.js, Express.js, SQLAlchemy, REST APIs, Authentication, JWT, RBAC
Frontend: React.js, Redux, Tailwind CSS
Databases: PostgreSQL, pgvector, MySQL, Redis, ChromaDB, Pinecone, FAISS
DevOps & Tools: Git, GitHub, Docker, AWS, CI/CD, Vercel, Render, Railway, Postman
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design Basics

EXPERIENCE

TechSonix Jan 2025 – Mar 2025
Backend Developer Intern Remote

- Developed and maintained Node.js/Express.js RESTful API endpoints using modular routing and middleware under senior developer guidance.
- Implemented JWT authentication, bcrypt password hashing and role-based access control (RBAC).
- Debugged backend issues, used Git, and collaborated on feature development and code reviews.

PROJECTS

DocMind – AI Document Intelligence Platform
Python, FastAPI, PostgreSQL/pgvector, Gemini/Groq APIs, React, Docker

- Built multi-user retrieval-augmented generation (RAG) document Q&A and summarization for PDF, DOCX and TXT using 768-dimensional Gemini embeddings and owner-filtered vector retrieval.
- Implemented claim-level citations, exact quote validation and a second-pass entailment check, with abstention when evidence checks fail.
- Built durable ingestion workers with bounded retries, crash recovery and atomic indexing/deletion in PostgreSQL; containerized the stack with Docker.
- Verified authentication, retrieval and failure recovery with 38 backend tests and 3 Chromium browser tests; achieved 78% backend code coverage.

ResolveDesk – Order Support Agent
Python, FastAPI, React.js, LangChain, LangGraph, LangSmith, PostgreSQL

- Built a tool-calling agent to investigate order issues, check return policies and propose resolutions; streamed tool activity and approval requests to React.
- Implemented checkpointed workflows, human approval before simulated refunds and idempotent actions to prevent duplicates on retries.
- Evaluated tool selection, policy compliance and approval bypass cases with LangSmith; traced latency, token usage and failed tool calls.

Investify – Financial Planning & Stock Research Platform | **Live Demo**
React.js, Node.js, Express.js, MongoDB, Upstox API, Docker

- Built a full-stack platform solo with risk-based asset allocation, financial goals, watchlists, SIP projections and PDF reports.
- Integrated Upstox for NSE/BSE quotes and MFapi for fund NAVs; built budget-aware stock screening with caching, rate limits, timeouts and stale-price checks.
- Secured APIs with revocable cookie sessions, bcrypt and CSRF protection; used MongoDB transactions and concurrency control to preserve plan history.
- Verified APIs, calculations and browser journeys with 37 automated tests; added Docker builds, CI and health checks.

EDUCATION

International Institute of Information Technology (I2IT), Pune Expected 2027
Bachelor of Engineering in Computer Engineering – CGPA: 7 Pune, Maharashtra

Sanjivani K.B.P. Polytechnic, Kopargaon 2024
Diploma in Computer Engineering – 90.29% Kopargaon, Maharashtra